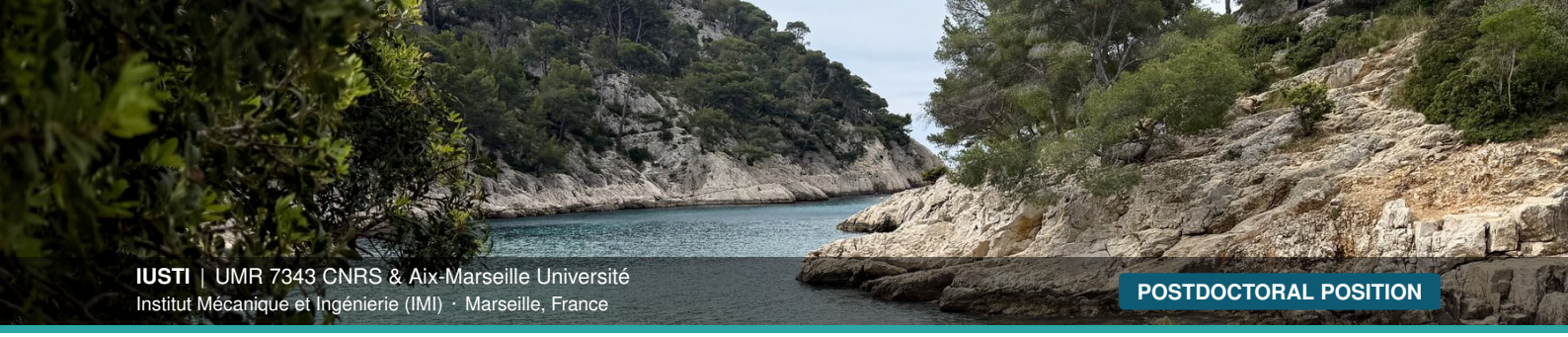


SNSF project *Hierarchy of Scales* · WP 1–3

Bridging the scales of soft fracture

Experiments linking slenderness, nonlinear elasticity, and dissipation at the crack tip

* * *

POSITION	Postdoctoral researcher, experimental	**DURATION**	Up to 3 years (project ends 2029)	**START**	As soon as possible
PARTNERS	ETH Zurich (D. Kammer, simulations) · CNRS Lyon (M. Adda-Bedia, theory) · IUSTI & IMI, Aix-Marseille Université				

The sub-project. Fracture of soft solids is a multiscale problem: the thickness of the body, the extent of the nonlinear-elastic zone near the tip (of order Γ/μ, the ratio of fracture energy to shear modulus), and the size of the dissipative region can all be comparable [1–4]. This postdoctoral position will be the experimental backbone of the project. The postdoc will lead the development and validation of the shared measurement methods: 3D particle tracking and strain-field reconstruction, J-integral and crack-tip opening analysis, and sample preparation for cryo-electron microscopy. The postdoc will also carry out experiments that link the three work packages, for instance by following how the near-tip fields and the fracture energy change as a crack evolves from a 3D, penny-shaped geometry to a through-crack in samples of decreasing thickness [5]. Working closely with the three PhD students of the project, the postdoc will be the main experimental interface with the simulation group at ETH Zurich and the theory group in Lyon, coordinating data formats, joint analysis sessions, and publications.

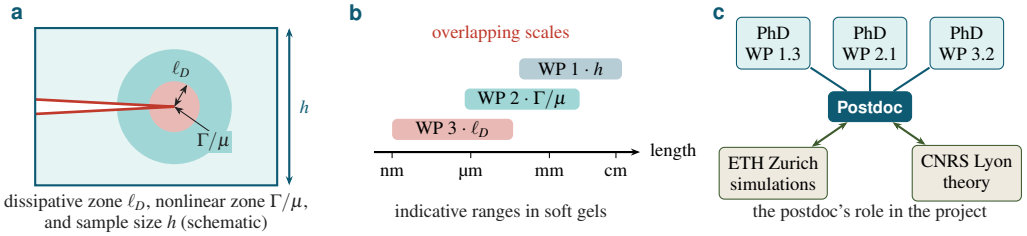


Figure 1. A hierarchy of scales (schematic). (a) Around the tip of a crack in a soft solid, a dissipative zone of size ℓ_D sits inside a nonlinear-elastic zone of size $\sim \Gamma/\mu$, both of which can be comparable to the sample thickness h. (b) The three work packages each target one of these scales, whose ranges overlap in soft gels. (c) The postdoc links the three PhD projects (WP 1.3, 2.1, 3.2) and the numerical and theoretical partners.

Context. This position is part of the SNSF project "The Hierarchy of Scales in the Fracture of Soft and Slender Bodies" (2025–2029). Whether and how a soft solid breaks depends on several length scales at once: the size of the body itself (its thickness, or slenderness), the nonlinear-elastic zone near the crack tip where the material is stretched far beyond linear response, and the region over which energy is dissipated by molecular processes. In soft materials these scales can be comparable, so classical fracture theories, which rely on their strict separation, break down. The project addresses each scale in a dedicated work package (WP) and connects them through combined experiments, simulations, and theory: experiments at Aix-Marseille Université, simulations in the group of Prof. D. Kammer at ETH Zurich, and analytical theory with Prof. M. Adda-Bedia (CNRS, Lyon). The experimental program is carried out at IUSTI (UMR 7343, CNRS / AMU) in the group of J. Kolinski, equipped with custom tensile loading devices, 3D light-sheet microscopy, and particle-tracking infrastructure, with access to AMU's cryo-electron microscopy facilities. IUSTI and the wider Institut Mécanique et Ingénierie (IMI) have strong expertise in rheology, granular and suspension mechanics, and solid mechanics; collaborations within the lab and across the IMI are strongly encouraged, as are research stays at ETH Zurich. The project will host an international workshop on soft fracture at Monte Verità (Ascona, Switzerland) in November 2027.

Candidate profile. We seek a candidate with a PhD in physics, mechanics, materials science, or a related field and a strong experimental record in fracture mechanics, soft matter, or experimental mechanics. Experience with 3D microscopy, particle tracking or digital image correlation, full-field analysis such as J-integral evaluation, or hydrogel synthesis is a clear asset, as is solid programming for data analysis (Python or MATLAB). Because the postdoc will help mentor three PhD students and coordinate exchanges across three institutions, we seek a candidate with strong communication and organizational skills and an interest in mentoring. The postdoc will also contribute to organizing the project's workshop at Monte Verità in November 2027. The working language of the group is English.

Contact and application. Send a CV, a one-page statement of interest, and the names of two referees to John Kolinski, john.kolinski@univ-amu.fr • kolinski.science. Applications will be reviewed until the position is filled.

[1] C. Creton and M. Ciccotti, "Fracture and adhesion of soft materials: a review," *Rep. Prog. Phys.* **79**, 046601 (2016).
[2] C.-Y. Hui, A. Jagota, S. J. Bennison, and J. D. Londono, "Crack blunting and the strength of soft elastic solids," *Proc. R. Soc. Lond. A* **459**, 1489–1516 (2003).
[3] E. Bouchbinder, T. Goldman, and J. Fineberg, "The dynamics of rapid fracture: instabilities, nonlinearities and length scales," *Rep. Prog. Phys.* **77**, 046501 (2014).
[4] R. Long and C.-Y. Hui, "Fracture toughness of hydrogels: measurement and interpretation," *Soft Matter* **12**, 8069–8086 (2016).
[5] X. Wei *et al.*, "Complexity of crack front geometry enhances toughness of brittle solids," *Nat. Phys.* **20**, 1009–1014 (2024).